

Amit Kumar

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Experience

Quansight (now OpenTeams)

Staff Software Engineer

May. 2020 – Present

Remote, London

- > Core maintainer of **Nebari**, open-source Kubernetes-native platform providing **cloud abstractions** over AWS EKS, GCP GKE, and Azure AKS, letting teams ship workloads without cloud-specific code; contributing to the next-gen rewrite in Go, OpenTofu, and ArgoCD.
- > Automated TLS provisioning (cert-manager), SSO (Keycloak/OIDC), and RBAC across multi-cloud K8s clusters.
- > Built **jhub-apps**, a JupyterHub extension that launches sandboxed Streamlit, Gradio and Panel apps on Kubernetes.
- > Managed cloud infrastructure across AWS, GCP, Azure and OpenStack (GPU servers with NVIDIA V100s, Ansible-based provisioning).
- > Led backend work on various client services (FastAPI, Flask, Django).
- > Optimised GPU kernels in population-genetics toolkit **sgkit**, delivering a **10x speed-up** (CUDA, Dask, RAPIDS).
- > Contributed to numerous Data Engineering, Data Science Infrastructure, and Backend client projects using tools in the PyData ecosystem (Dask, Numba, RAPIDS).

Cirun.io

Founder & Principal Engineer

June 2020 – Present

Remote

- > Built **core infrastructure** for a multi-cloud CI/CD platform; a **cloud abstraction layer** over **7 providers** (AWS, GCP, Azure, OCI, DigitalOcean, OpenStack, Vast.ai) that dynamically provisions ephemeral VMs via native APIs and Terraform.
- > Powers GPU CI for **conda-forge**, one of the largest open-source package ecosystems.
- > Implemented observability with **OpenTelemetry** (Flask + Celery instrumentation, OTLP export) and **Beszel**; led 24x7 on-call rotation sustaining **99.99%** uptime.
- > Tech Stack: Python (Flask, Celery), Postgres, Redis, Terraform, Cloudflare Workers, GitHub Actions API.

Contract Software Engineer

Software Engineer

June. 2018 – Mar. 2020

London, UK

- > Built backend services and APIs for Aire Labs (Fintech), Turner & Townsend, and Cambridge Medical Robotics.
- > Designed data pipelines with Airflow; wrote microservices on AWS Lambda for credit assessment.
- > Designed and implemented APIs to make construction data accessible to the business (Turner & Townsend).
- > Built system testing infrastructure for surgical robot; responsible for code review, mentoring and diagnosing software issues (Cambridge Medical Robotics).

The Telegraph

Software Engineer, Data Platform

Feb. 2018 – May. 2018

London, UK

- > Designed ETL pipelines; improved deployment infrastructure with Jenkins.
- > Led Python 2 to 3 migration for ETL pipeline codebase.

Powervault

Software Engineer, Backend

Oct. 2017 – Feb. 2018

London, UK

- > Integrated external services to control IoT devices and improved client-facing dashboards.
- > Dockerized backend stack; implemented CI/CD on GitLab.

Zomato

Software Development Engineer

Oct. 2016 – Oct. 2017

Gurgaon, India

- › Improved user credibility system and rewrote rule-based spam detection system.
- › Built data pipeline for image quality prediction and a cuisine predictor using word embeddings.
- › Created internal dashboards for analytics and visualization.

Google Summer of Code 2015-2020

Python Software Foundation, SymPy

Summer 2015

Remote

- › Core contributor to SymPy equation solvers; implemented Complex Sets and linear system solver (GSoC 2015, Python Software Foundation).
- › Mentored GSoC students 2016–2020.

Education

Delhi Technological University

Bachelor of Technology in Mathematics and Computing Engineering

2012 – 2016

New Delhi, India

Presentations

20+ conference talks across 8 countries (PyCon, SciPy, JupyterCon, PyData), iamit.in/talks

Skills

Languages Python, Go, JavaScript, Bash, SQL, C/C++ (familiar)

Infrastructure & Cloud Kubernetes (EKS, GKE, AKS) at scale, Terraform, OpenTofu, Helm, ArgoCD, Docker, Ansible, AWS, GCP, Azure, OpenStack, Cloudflare Workers

Backend & Observability Flask, FastAPI, Django, Celery, PostgreSQL, Redis, OpenTelemetry, Keycloak, cert-manager, Nginx, Traefik

Honors & Awards

Exceptional Talent in Technology (Endorsement) by Tech City UK (Government of UK)

Financial Aid Award Presentation on SymPy at SciPy 2016

SemiFinal HackerRank WorldCup coding competition (Team size: 2)

Open Source Contributions

- Nebari (Core Developer), SymPy (Core Developer), conda-forge, Panel, sgkit, pyflyby, jhub-apps, SymEngine, SymPy-Gamma, Papyri, Dask Gateway, NumPy, Flask-Caching, CPython, pandas-gbp, facebook-prophet, pydsa